

SAT Math Mastery · Week 4A

Quadratics II: Discriminant, Tangency & Vieta

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics number of solutions via the discriminant · tangent line to a parabola · sum and product of roots · nonlinear systems (line + parabola, line + circle) · parameter conditions · identities true for all x

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Decide the number of real solutions of $ax^2 + bx + c = 0$ from $\Delta = b^2 - 4ac$ (also from a graph).
- S2.** Find a constant that makes an equation have exactly one, two, or no real solutions (strict vs. non-strict inequalities, integer counts).
- S3.** Use Vieta: sum $-\frac{b}{a}$ and product $\frac{c}{a}$ of the solutions without solving; build symmetric expressions such as $r^2 + s^2$.
- S4.** Substitute a line into a parabola or circle, then use Δ to count intersection points or force tangency.
- S5.** Treat “intersect at two points” problems with the sum and product of the x -coordinates (midpoint, distance).
- S6.** Handle hidden conditions: leading coefficient could be 0, “distinct” vs. “real”, sign conditions on roots.
- S7.** Match coefficients when two expressions are equal for all x (identities).
- S8.** Read the sign of a , b , c and Δ from the picture of a parabola.

2 · REQUIRED FACTS

- $\Delta > 0$: two distinct real solutions. $\Delta = 0$: exactly one real (repeated) solution. $\Delta < 0$: no real solutions.
- A line is tangent to a parabola (or circle) exactly when the substituted quadratic has $\Delta = 0$: one common point.
- For $ax^2 + bx + c = 0$ with solutions r and s : $r + s = -\frac{b}{a}$ and $rs = \frac{c}{a}$. These hold whether or not the solutions are nice.
- “No real solutions” $\iff$ the graph of $y = ax^2 + bx + c$ never touches the x -axis.
- If a and c have opposite signs, then $\Delta > 0$ (the parabola must cross the x -axis).
- If the leading coefficient contains a parameter, check the value that makes it 0: the equation stops being quadratic.

3 · REQUIRED FORMULAS

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad \Delta = b^2 - 4ac, \quad |r - s| = \frac{\sqrt{\Delta}}{|a|}$$

$$r + s = -\frac{b}{a}, \quad rs = \frac{c}{a}, \quad r^2 + s^2 = (r + s)^2 - 2rs, \quad r^3 + s^3 = (r + s)^3 - 3rs(r + s)$$

$$\text{vertex: } x = -\frac{b}{2a}, \quad (x - r)(x - s) = x^2 - (r + s)x + rs, \quad \text{one solution } \iff ax^2 + bx + c = a(x - h)^2$$

4 · MINI THEORY SHEET

Count solutions in three steps. (1) Move everything to one side: $ax^2 + bx + c = 0$. (2) Compute $\Delta = b^2 - 4ac$ (keep the constant k symbolic). (3) Turn the wording into an inequality on Δ .

Wording	Condition on Δ
exactly one (real) solution, tangent, touches	$\Delta = 0$
two distinct real solutions, crosses twice	$\Delta > 0$
no real solutions, never crosses / meets	$\Delta < 0$
at least one real solution	$\Delta \geq 0$
the <i>greatest integer</i> with no solution	largest integer satisfying the strict inequality

Line meets curve. Substitute the line $y = mx + k$ into the curve. The result is a quadratic in x whose Δ counts intersection points. Tangency $\iff \Delta = 0$.

Two intersection points. The x -coordinates are the two solutions, so their sum and product come straight from Vieta; the y -coordinates follow from the line.

Parameter in the leading coefficient. For $(k-1)x^2 + \dots = 0$, test $k = 1$ separately: the equation becomes linear and may have exactly one solution.

Identities. If two expressions are equal for every x , expand and match the coefficients of x^2 , x and the constants.

Reading a graph. Opens up $\Rightarrow a > 0$. Vertex right of the y -axis $\Rightarrow -\frac{b}{2a} > 0$, so a and b have opposite signs. y -intercept = c . Crosses the x -axis twice $\Rightarrow \Delta > 0$.

Speed shortcuts.

Situation	Shortcut
sum or product of solutions asked $r^2 + s^2$, $r^3 + s^3$, $\frac{r}{s} + \frac{s}{r}$	Vieta: no solving, no factoring; first rewrite as $ax^2 + bx + c = 0$ express with $r + s$ and rs
$x^2 + bx + c$ with exactly one solution	$c = \left(\frac{b}{2}\right)^2$ (perfect square)
$ r - s $ between two solutions	$\frac{\sqrt{\Delta}}{ a }$
line $y = mx + k$ and $y = x^2 + \dots$	set equal, one quadratic, then Δ
circle $x^2 + y^2 = R^2$ and line	line tangent $\iff$ distance from centre to line = R
number of integer k with $\Delta < 0$	$k^2 < N$ gives $-\sqrt{N} < k < \sqrt{N}$: count the integers carefully (strict)
$ax^2 + bx + c = 0$ with $ac < 0$	always two real solutions

5 · GUIDED EXAMPLES

Example 1 — Constant that gives exactly one solution

For what value of k does $3x^2 - 12x + k = 0$ have exactly one real solution?

$\Delta = (-12)^2 - 4(3)(k) = 144 - 12k$. Exactly one solution: $\Delta = 0 \Rightarrow k = 12$. (Check: $3x^2 - 12x + 12 = 3(x-2)^2$.)

Example 2 — Tangent line to a parabola

The line $y = 4x + k$ is tangent to the parabola $y = x^2 + 2x + 5$. Find k .

Set equal: $x^2 + 2x + 5 = 4x + k \Rightarrow x^2 - 2x + (5 - k) = 0$. Tangent $\Rightarrow \Delta = 0$: $4 - 4(5 - k) = 0 \Rightarrow k = 4$.
(The touching point: $x = 1$, $y = 8$.)

Example 3 — Vieta without solving

The solutions of $2x^2 - 9x + 3 = 0$ are r and s . Find $r^2 + s^2$.

$r + s = \frac{9}{2}$, $rs = \frac{3}{2}$. Then $r^2 + s^2 = (r + s)^2 - 2rs = \frac{81}{4} - 3 = \frac{69}{4}$.

6 · PRACTICE SET — 46 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

How many distinct real solutions does the equation $2x^2 - 5x + 4 = 0$ have?

- (A) Infinitely many (B) Zero
(C) Exactly one (D) Exactly two

2. EASY · NO CALCULATOR NEEDED

What is the sum of the solutions of $x^2 - 9x + 14 = 0$?

3. EASY · NO CALCULATOR NEEDED

What is the product of the solutions of $3x^2 - 7x - 12 = 0$?

- (A) $\frac{7}{3}$ (B) $-\frac{7}{3}$
(C) -4 (D) 4

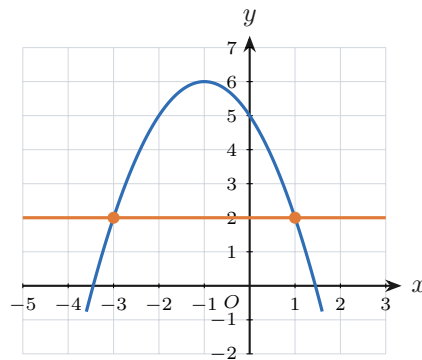
4. EASY · NO CALCULATOR NEEDED

The equation $x^2 + 6x + c = 0$ has exactly one real solution, where c is a constant. What is the value of c ?

5.

EASY · DESMOS

The graph of a system of two equations in the xy -plane is shown. How many solutions does the system have?



(A) 2

(B) 3

(C) 0

(D) 1

6.

MEDIUM · NO CALCULATOR NEEDED

For which of the following values of k does the equation $2x^2 + kx + 8 = 0$ have no real solutions?

(A) -9 (B) -8 (C) 8 (D) 5

7.

MEDIUM · NO CALCULATOR NEEDED

The equation $2x^2 + 8x + k = 0$ has no real solutions, where k is an integer. What is the least possible value of k ?

8.

MEDIUM · NO CALCULATOR NEEDED

The equation $3x^2 - 5 = 11x + x^2$ has two real solutions. What is the sum of the solutions?

(A) $-\frac{11}{2}$

(B) $-\frac{5}{2}$

(C) $\frac{11}{2}$

(D) 11

9.

MEDIUM · NO CALCULATOR NEEDED

In the xy -plane, the line $y = 2x + k$ is tangent to the parabola $y = x^2 + 4x + 7$, where k is a constant. What is the value of k ?

(A) 3

(B) 6

(C) 7

(D) -6

10.

MEDIUM · DESMOS

In the xy -plane, the line $y = 6$ intersects the parabola $y = x^2 - 6x + 11$ at two points. What is the distance between the two points?

(A) 5

(B) 16

(C) 2

(D) 4

11.

MEDIUM · DESMOS

$$y = x^2 - 5$$

$$y = 2x + 3$$

The solutions to the given system of equations are (x_1, y_1) and (x_2, y_2) . What is the value of $y_1 + y_2$?

(A) 10

(B) 12

(C) 2

(D) 8

12.

MEDIUM · NO CALCULATOR NEEDED

The solutions of $x^2 - 10x + c = 0$ are r and s , where $r > s$ and $r - s = 4$. What is the value of c ?

13.

MEDIUM · NO CALCULATOR NEEDED

A rectangular garden has a perimeter of 36 meters and an area of 80 square meters. What is the length, in meters, of the longer side of the garden?

14.

HARD · NO CALCULATOR NEEDED

In the equation $5x^2 + (k - 3)x + 5 = 0$, k is a constant. For which of the following values of k does the equation have exactly one real solution?

(A) -10 (B) -7 (C) 7 (D) 10 **15.**

HARD · NO CALCULATOR NEEDED

In the xy -plane, the line $y = mx - 5$, where $m > 0$, is tangent to the parabola $y = 2x^2 + 3$. What is the value of m ?

16.

HARD · NO CALCULATOR NEEDED

The solutions of $x^2 - 7x + 5 = 0$ are r and s . What is the value of $r^2 + s^2$?

(A) 29 (B) 59 (C) 39 (D) 49

17.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the line $y = x + k$, where $k > 0$, is tangent to the circle $x^2 + y^2 = 50$. What is the value of k ?

18.

HARD · NO CALCULATOR NEEDED

For how many integer values of k does the equation $3x^2 + kx + 12 = 0$ have no real solutions?

(A) 23

(B) 24

(C) 11

(D) 22

19.

HARD · NO CALCULATOR NEEDED

The equation $x^2 - 9x + k = 0$ has two real solutions, and one solution is twice the other. What is the value of k ?

20.

HARD · NO CALCULATOR NEEDED

In the equation $ax^2 + bx + c = 0$, a , b , and c are constants with $a > 0$ and $c < 0$. Which of the following must be true?

I. The equation has two distinct real solutions.

II. The two solutions are both positive.

III. The product of the two solutions is negative.

(A) I, II, and III

(B) I only

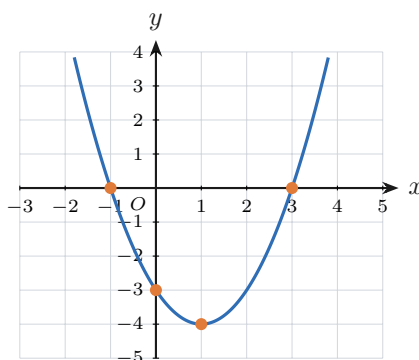
(C) III only

(D) I and III only

21.

HARD · DESMOS

The graph of $y = f(x)$ is shown in the xy -plane. For which of the following values of k does the equation $f(x) + k = 0$ have no real solutions?



(A) -4

(B) 4

(C) 5

(D) -5

22.

HARD · CALCULATOR

A ball is thrown upward from a platform. Its height, in feet, t seconds after it is thrown is $h(t) = -16t^2 + 64t + 6$. The ball is at a height of exactly H feet at exactly one time during its flight, where $H > 6$. What is the value of H ?

23.

HARD · NO CALCULATOR NEEDED

The equation $(ax + 3)(2x - b) = 6x^2 + cx - 12$ is true for all values of x , where a , b , and c are constants. What is the value of c ?

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a constant k , the equation $(k - 1)x^2 - 6x + (k - 1) = 0$ has exactly one real solution. What is the sum of all possible values of k ?

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

The solutions of $x^2 - 6x + 4 = 0$ are r and s . What is the value of $r^3 + s^3$?

- (A) 108 (B) 144
(C) 216 (D) 72

26.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $y = x + 2$ intersects the circle $x^2 + y^2 = 20$ at points P and Q . What is the value of $(PQ)^2$?

27.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $y = 2x - 3$ is tangent to the parabola $y = x^2 - 6x + c$, where c is a constant. The point of tangency is (p, q) . What is the value of $p + q$?

(A) 5

(B) 13

(C) 4

(D) 9

28.

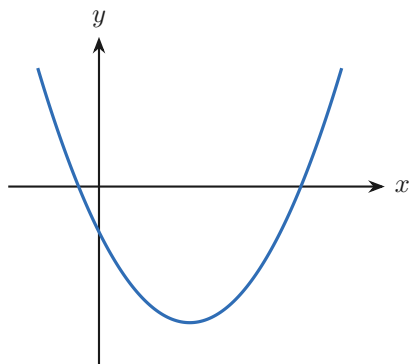
SAT CHALLENGE · NO CALCULATOR NEEDED

The graph of $y = ax^2 + bx + c$ is shown in the xy -plane, where a , b , and c are constants. Which of the following must be true?

I. $b^2 - 4ac > 0$

II. $ab < 0$

III. $c > 0$



(A) I and II only

(B) II and III only

(C) I, II, and III

(D) I only

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

What is the least integer k for which the equation $x^2 - kx + (k + 3) = 0$ has two distinct positive real solutions?

(A) 6

(B) 7

(C) 8

(D) 3

30.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $y = 2x + k$ intersects the parabola $y = x^2 - 4x + 1$ at two points whose x -coordinates differ by 8. What is the value of k ?

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

For real numbers p and q with $p^2 + 4q^2 = 72$, the product $(2x + p)(x + q)$ is equivalent to $2x^2 + bx + c$, where b and c are constants. What is the greatest possible value of c ?

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the line $y = mx - 5$ intersects the parabola $y = x^2 - 6x + 11$ at exactly one point, where m is a negative constant. What is the value of m ?

(A) 2

(B) 14

(C) -14

(D) -2

33.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $x + y = 4$ intersects the circle $x^2 + y^2 - 4x + 2y = 24$ at two points. What is the distance between these two points?

(A) $7\sqrt{2}$

(B) 14

(C) $\frac{7\sqrt{2}}{2}$

(D) 7

34.

SAT CHALLENGE · DESMOS

$$x^2 + y^2 = 65$$

$$x - y = 7$$

The solution (a, b) to the system of equations above satisfies $a > 0$. What is the value of $a + 3b$?

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

For integers a , b , c , and d , $(ax + b)(cx + d) = 6x^2 + kx + 5$ for all values of x , where k is a constant. Which of the following must be true?

I. k is oddII. When k is divided by 4, the remainder is 1III. k is not divisible by 5

(A) II and III only

(B) I, II, and III

(C) I only

(D) I and III only

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the line $y = k(x - 1) - 2$, where k is a positive constant, is tangent to the parabola $y = x^2 + 2x + 4$. What is the value of k ?

(A) 4

(B) 8

(C) 10

(D) -2

37.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the graph of the quadratic function p and the graph of the line $y = 5 - x$ intersect at exactly two points, one with x -coordinate 1 and one with x -coordinate 4. If $p(0) = -3$, what is the value of $p(6)$?

38.

SAT CHALLENGE · DESMOS

In the xy -plane, the graphs of $y = x^2 + 6x + 4$ and $y = mx - 4$ have no point in common, where m is a constant. What is the sum of all integer values of m ?

39.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$(x + p)(x + q) = x^2 + bx + c$$

In the equation above, p and q are real constants and b and c are integers. Which of the following must be true?

I. $p + q$ is an integer

II. $p^2 + q^2$ is an integer

III. $\frac{p^2 + q^2}{2}$ is an integer

(A) I only

(B) I and II only

(C) II and III only

(D) I, II, and III

40.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^2 + y^2 = 45$$

$$y = mx + 15$$

In the system above, m is a negative constant. If the system has exactly one solution (x, y) , what is the value of x ?

(A) -6

(B) -3

(C) 3

(D) 6

41.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$y = 3x^2 + 4$$

$$y = 1 - kx$$

In the system above, k is a constant. For which of the following values of k does the system have two distinct solutions (x, y) whose x -coordinates are both negative?

(A) 7

(B) -7

(C) 2

(D) 6

42.

SAT CHALLENGE · CALCULATOR

In the xy -plane, the line $y = mx + 10$, where m is a constant, intersects the circle $x^2 + y^2 = 12$ at exactly one point. What is the value of m^2 ?

43.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the line $y = kx + 4$, where k is a constant, is tangent to the parabola $y = x^2 + 4x + 13$ for exactly two values of k . For each of these values the line touches the parabola at one point. What is the greatest y -coordinate of these two points of tangency?

(A) 13

(B) 34

(C) 3

(D) 10

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function $g(x) = 2x^2 + bx + c$, where b and c are constants, has zeros $k + 1$ and $3k - 5$, where k is a constant. If $g(2) = g(10)$, what is the value of c ?

(A) 24

(B) 35

(C) 70

(D) 140

45.

SAT CHALLENGE · NO CALCULATOR NEEDED

For nonzero real numbers p and q , the equation $px^2 + 3qx + 4p = 0$ has exactly one real solution, and $pq = -48$. If $p > 0$, what is the value of $p + q$?

(A) -2 (B) 2 (C) 14 (D) -14

46.

SAT CHALLENGE · NO CALCULATOR NEEDED

Positive real numbers x and y satisfy $x \neq y$, $x + y = 15$ and $xy = 36$. What is the value of

$$\frac{\frac{1}{x} - \frac{1}{y}}{\frac{1}{x^2} - \frac{1}{y^2}} ?$$

(A) $\frac{5}{12}$ (B) 1 (C) 4 (D) $\frac{12}{5}$